Shubhadeep Mandal

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Education

International Institute of Information Technology, Hyderabad July 2025 – Present
Telangana, India
Master of Technology in Computer Science and Engineering *CGPA: 9.12*

Indian Institute of Information Technology, Ranchi Aug 2020 – Jun 2024
Ranchi, India
Bachelor of Technology (Hons.) in Computer Science and Engineering *CGPA: 9.05*

Projects

Negotiate, Then Solve: LLM Agents for Timetabling | *Python, Gemini, Qwen, RAG, CP-SAT, FastAPI*

- Built an LLM assistant that turns scheduling requests into constraints for a solver and negotiates conflicts with people.
- Added safety checks (authority, cited rules, human approval); **no attack succeeded** in 50 prompt-injection tests.
- Built hybrid keyword and embedding search over policy documents, reaching **93%** recall.
- Fine-tuned a small 4B model with LoRA that runs on a laptop GPU and outperformed the Gemini model it learned from.

IndicLLM: Transformer Pretraining & Reasoning SFT | *PyTorch, SentencePiece, ONNX Runtime Web*

- Pretrained GPT-style language models for Hindi and Assamese from scratch in PyTorch, including the data and tokenizers.
- Fine-tuned them for multi-step reasoning, reaching **96%** test accuracy against a 48% baseline.
- Tested whether the models were reasoning or memorising answers before reporting results.
- Exported the models to run fully in the browser and deployed them as a live demo.

Retrieve Ten, Keep Three: Re-ranking RAG Evaluation | *Python, FAISS, BM25, FastAPI*

- Built a question-answering system over the GDPR that retrieves 10 passages and re-ranks them to keep the best 3.
- Re-ranking raised top-3 retrieval accuracy from **46% to 85%**; combining keyword and embedding search helped further.
- Built an LLM judge checked against hand labels, and found many "wrong" answers came from mislabelled test data.

Football Video Analytics with YOLO | *Python, YOLOv8, OpenCV, Streamlit*

- Built a video analysis pipeline that tracks players, referees and the ball with a fine-tuned YOLOv8 model.
- Grouped players into teams by jersey colour and mapped the camera view to the pitch to measure speed and distance.
- Streamed frames in batches to cut memory use from **5.3 GB to 1 GB**, and built a Streamlit app to explore the results.

Technical Skills

Languages: Python, C/C++, TypeScript, JavaScript, SQL, Bash.
ML & LLMs: PyTorch, Hugging Face, LoRA, RAG, FAISS, llama.cpp, Gemini API.
Data & Vision: NumPy, Pandas, Scikit-learn, OpenCV, YOLOv8.
Tools: FastAPI, Streamlit, Linux, Git, Docker, pytest.

Achievements

- **B.Tech Rank Holder:** Ranked 2nd by GPA in the B.Tech CSE batch at IIIT Ranchi.
- **PGEE 2025:** Secured All-India Rank 100 in the IIIT Hyderabad Post Graduate Entrance Examination.
- **Megathon 2025:** Placed 3rd in the mobile computing track.

Interests

- Reading web novels and manga.
- Playing chess and badminton.